

*An Investigation of Graph Neural Networks
for Molecular Property Prediction*

Eoghan Stafford

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Schrödinger Equation

$$\hat{H}\Psi(\{\vec{R}_A\}, \{\vec{r}_i\}) = E\Psi(\{\vec{R}_A\}, \{\vec{r}_i\})$$

$$\hat{H} = -\sum_{i=1}^{N_e} \frac{1}{2} \nabla_i^2 - \sum_{A=1}^{N_A} \frac{1}{2M_A} \nabla_A^2 - \sum_{i,A} \frac{Z_A}{|\vec{r}_i - \vec{R}_A|} + \sum_{i < j} \frac{1}{|\vec{r}_i - \vec{r}_j|} + \sum_{A < B} \frac{Z_A Z_B}{|\vec{R}_A - \vec{R}_B|}$$

DFT

Maps problem to
electron density

Accuracy:

Medium

Cost:

$\mathcal{O}(N^3)$, Hours

Hartree-Fock

Electrons see average
field of others

Accuracy:

Low

Cost:

$\mathcal{O}(N^4)$, Hours

CCSD(T)

Correlated wave
function method

Accuracy:

High

Cost:

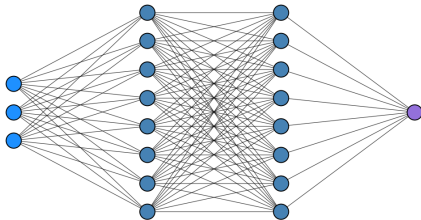
$\mathcal{O}(N^7)$, Days

- 1 Generate training data
DFT, CCSD(T)
- 2 Train ML model
- 3 Instant predictions
- 4 Further chemical space exploration

How Neural Networks Work

Multi-feature input $\rightarrow$ Column vector

Singular feature input $\rightarrow$ Scalar



$$a^{(t+1)} = \sigma(W^{(t)}a^{(t)} + b^{(t)})$$

Single composed function of entire network:

$$\hat{y} = \sigma \left(W^{(n)} \sigma \left(W^{(n-1)} \sigma \left(\dots \sigma \left(W^{(1)} \vec{x} + \vec{b}^{(1)} \right) \right) + \vec{b}^{(n-1)} \right) + \vec{b}^{(n)} \right)$$

Training: Backpropagation

Loss Function (MSE)

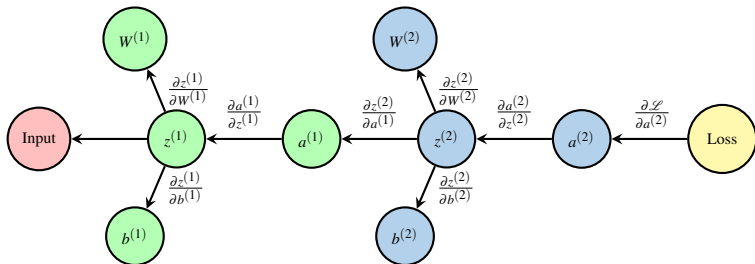
$$\mathcal{L}(\hat{y}, y) = \frac{1}{2}(\hat{y} - y)^2$$

Gradient Descent Update

$$W_{ik}^{(1)} \leftarrow W_{ik}^{(1)} - \eta \frac{\partial \mathcal{L}}{\partial W_{ik}^{(1)}}$$

η = learning rate

Chain rule:
$$\frac{\partial \mathcal{L}}{\partial W_{ik}^{(1)}} = \left(\frac{\partial \mathcal{L}}{\partial z^{(n)}} \right) \left(\frac{\partial z^{(n)}}{\partial a^{(n-1)}} \right) \cdots \left(\frac{\partial a^{(1)}}{\partial z^{(1)}} \right) \left(\frac{\partial z^{(1)}}{\partial W_{ik}^{(1)}} \right)$$



Graph Neural Networks

Permutation Invariance

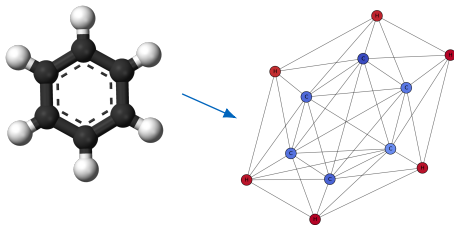
Order of atoms doesn't affect prediction.
Essential for molecules.

Spatial Encoding

Naturally captures 3D structure via distances and angles between atoms.

Message Passing

Information propagates through graph.
Each atom learns from its neighborhood.



Benzene molecule to GNN

Embedding



$$E(Z) \rightarrow \vec{x}_i \in \mathbb{R}^d$$

Atomic representations

Two-Body

$$d_{ij} = \|\vec{r}_i - \vec{r}_j\| < 3.3\text{\AA}$$

$$\text{rbf}_{ij} = \exp(-\gamma(d_{ij} - \mu_k)^2)$$

RBF expansion + message passing

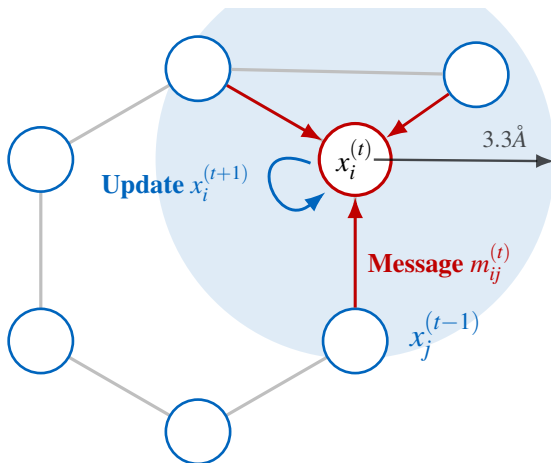
Three-Body

$$\cos \theta_{jik} = \frac{\vec{r}_{ij} \cdot \vec{r}_{ik}}{d_{ij} \cdot d_{ik}}$$

$$\text{sbf}_{ij} = \{P_l(\cos \theta_{jik})\}_{l=1}^N$$

Angular encoding (Legendre)

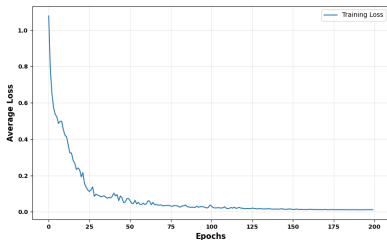
Message Passing



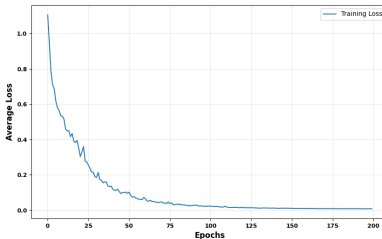
$$m_{ij}^{(t)} = h_{ij} \odot x_j^{(t)}, \quad h_{ij} = \text{MLP}(e_{ij}) \quad \left| \quad x_i^{(t+1)} = x_i^{(t)} + \sum_{j \in \mathcal{N}(i)} m_{ij}^{(t)} \left(x_i^{(t)}; x_j^{(t)}, e_{ij} \right)$$

Results: Model Training

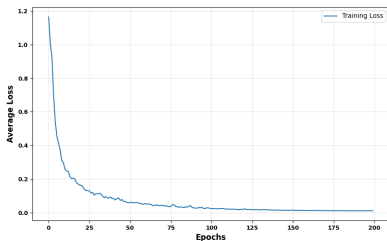
Benzene



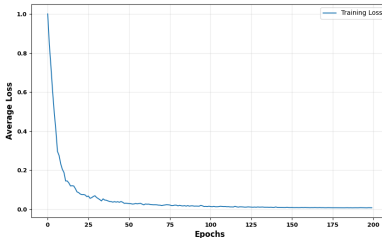
Aspirin



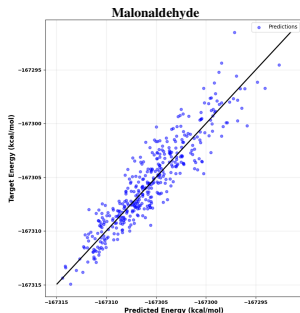
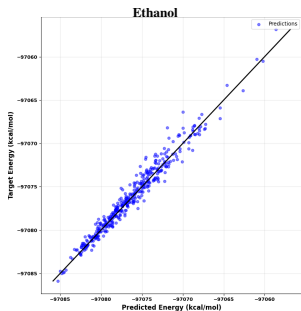
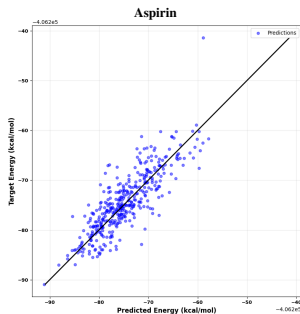
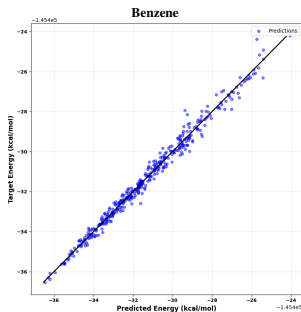
Malonaldehyde



Ethanol



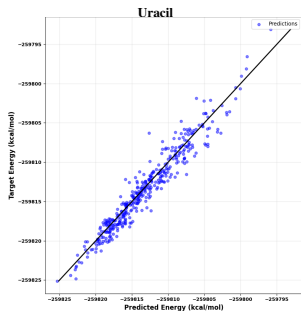
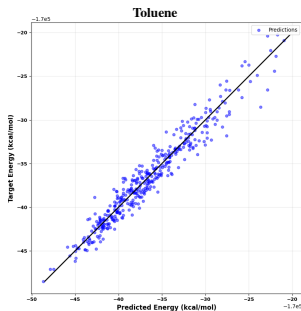
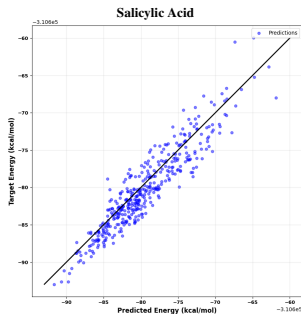
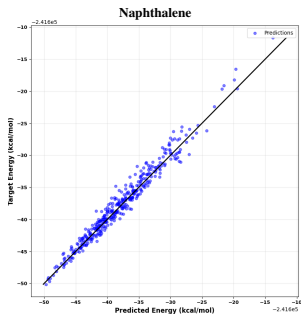
Results: Energy Prediction



Hyperparameters

- Learning rate:
 - Min: 1×10^{-7}
 - Max: 1×10^{-3}
- Weight Decay: 1×10^{-4}
- Training set: 1600
- Testing set: 400
- Batch size: 32
- Epochs: 200
- Activation: SiLU
- Hidden dim: 128
- Message layers: 3
- Dropout: 0.1

Results: Energy Prediction



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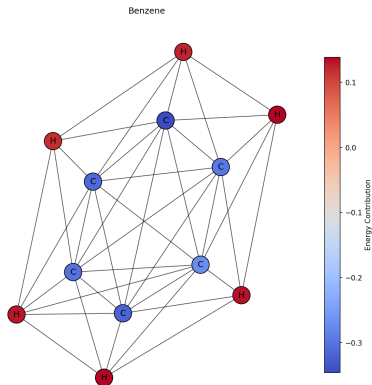
Results: Model Performance

Molecule	Train Loss	Test Loss	R^2	MAE	RMSE
Benzene	0.0117	0.0144	0.9861	0.2059	0.2841
Ethanol	0.0063	0.0389	0.9630	0.6258	0.8335
Aspirin	0.0083	0.3045	0.6916	2.5535	3.4218
Malonaldehyde	0.0112	0.1513	0.8399	1.2798	1.6225
Uracil	0.0115	0.0665	0.9338	0.9950	1.3212
Salicylic acid	0.0219	0.1365	0.8508	1.6745	2.0839
Naphthalene	0.0165	0.0350	0.9646	0.8412	1.0949
Toluene	0.0185	0.0547	0.9406	0.9565	1.2316

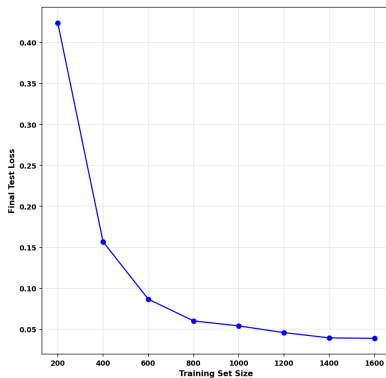
R^2 - Coefficient of determination: A number between 0 and 1 that measures how well a statistical model predicts an outcome.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

Results: Model Performance



Atom contributions to final energy in Benzene



Training set size vs. Final test loss for Ethanol

Conclusions

- Our three-body GNN achieves a lower MAE than SchNet (two-body GNN) on benchmark molecules, demonstrating the value of angular encoding.
- Machine learning enables large scale molecular screening. Once trained, our model predicts energies instantly enabling further exploration of chemical space.
- Unlike standard neural networks, GNNs operate on sparse local neighbourhoods around each node which is an essential property for molecular predictions.

Molecule	This Work	SchNet
Benzene	0.21	1.19
Ethanol	0.63	0.93
Aspirin	2.55	4.20
Malonaldehyde	1.28	2.03
Uracil	1.00	2.26
Salicylic Acid	1.67	3.27
Napthalene	0.84	3.58
Toluene	0.96	2.95

Test MAE (kcal/mol)

Thank You

Questions?

1. K. T. Shutt, P. J. Kindermans, H. E. Sauceda, S. Chmiela, A. Tkatchenko, *SchNet – A deep learning architecture for molecules and materials*, J. Chem. Phys. 148, 241722, 2018.
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2. Aaron Courville, Ian Goodfellow, and Yoshua Bengio , *Deep Learning*, MIT Press, 2016.
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3. Author(s), *Programming PyTorch for Deep Learning*, O'Reilly, 2019.
[PyTorch for Deep Learning](#)
4. Anders S. Christensen, Anatole Von lilienfeld, *Revised MD17 Dataset*, 2020.
[rMD17 Dataset](#)
5. Oliver Wiedera, Stefan Kohlbachera, Mélaïne Kuenemannb, Arthur Garona, Pierre Ducrota, Thomas Seidela, Thierry Langer, *A compact review of molecular property prediction with graph neural networks*, 2020.
[A compact review of molecular property prediction with graph neural networks](#)

Message Passing

